



Dresden Nuclear Power Station
6500 North Dresden Road
Morris, IL 60450

SVPLTR # 22-0046

10 CFR 50.73

October 28, 2022

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Dresden Nuclear Power Station, Unit 2
Renewed Facility Operating License No. DPR-19
NRC Docket No. 50-237

Subject: Licensee Event Report 237/2022-001-01, Reactor Scram due to Turbine Trip on High Reactor Water Level caused by Feedwater Regulating Valve Failure

Enclosed is Licensee Event Report 237/2022-001-01, Reactor Scram due to Turbine Trip on High Reactor Water Level caused by Feedwater Regulating Valve Failure. This report describes an event being reported in accordance with 10 CFR 50.73(a)(2)(iv)(A) an event that resulted in automatic actuation of a system listed in paragraph 10 CFR 50.73(a)(2)(iv)(B).

There are no regulatory commitments contained in this submittal.

Should you have any questions concerning this letter, please contact Duane Avery, Acting Regulatory Assurance Manager, at (815) 416-2804.

Respectfully,

A handwritten signature in blue ink, appearing to read "Patrick J. Boyle".

Patrick J. Boyle
Site Vice President
Dresden Nuclear Power Station

Enclosure: Licensee Event Report 237/2022-001-01

CC: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – Dresden Nuclear Power Station

NRC FORM 366 (08-2020)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104		EXPIRES: 08/31/2023	
 LICENSEE EVENT REPORT (LER) (See Page 3 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)							
1. Facility Name Dresden Nuclear Power Station, Unit 2				2. Docket Number 05000237		3. Page 1 OF 3	
4. Title Unit 2 Reactor Scram due to Turbine Trip on High Reactor Water Level caused by Feedwater Regulating Valve Failure							
5. Event Date			6. LER Number			7. Report Date	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day
07	12	2022	2022	001 -	01	10	28
						8. Other Facilities Involved	
						Facility Name N/A	
						Docket Number N/A	
9. Operating Mode 1						10. Power Level 100	
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)							
☐ 10 CFR Part 20		☐ 20.2203(a)(2)(vi)		☐ 50.36(c)(2)		☒ 50.73(a)(2)(iv)(A)	
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)	
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)	
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)	
☐ 20.2203(a)(2)(i)		☐ 20.2203(a)(2)(i)		☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(v)(D)	
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)	
☐ 20.2203(a)(2)(iii)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)	
☐ 20.2203(a)(2)(iv)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)		☐ 50.36(c)(1)(iii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)	
☐ OTHER (Specify here, in abstract, or NRC 366A).							
12. Licensee Contact for this LER							
Licensee Contact Duane Avery – Acting Regulatory Assurance Manager						Phone Number (Include area code) 815-416-2804	
13. Complete One Line for each Component Failure Described in this Report							
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component
X	SJ	SOL	A499	Y			
					Cause	System	Component
14. Supplemental Report Expected					15. Expected Submission Date		
☒ No					☐ Yes (If yes, complete 15. Expected Submission Date)		
					Month	Day	Year
16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)							
On July 12, 2022, at 0803 CDT, an automatic reactor protection system actuation occurred following a main turbine trip due to high reactor water level. The safety significance of this event was minimal. All control rods inserted to their full-in position. Following the reactor scram, all systems operated as expected.							
The cause of high reactor water level is due to a feedwater regulating valve unexpectedly failing full-open. The root cause of the feedwater regulating valve failing full-open was due to the feedwater regulating valve pilot valve assembly not operating due to a manufacturing byproduct inside the pilot valve introduced during the manufacturing process.							
Corrective actions included replacement of the respective air solenoid valve. Additional corrective actions include inspecting and certifying these incoming valves are free of foreign material. Also, the solenoid valves for the feedwater regulating valves were reclassified as single point vulnerabilities.							
This event is being reported in accordance with 10 CFR 50.73(a)(2)(iv)(A), an event that resulted in automatic actuation of a system listed in paragraph 10 CFR 50.73(a)(2)(iv)(B), specifically the reactor protection system including reactor scram.							

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Dresden Nuclear Power Station, Unit 2	05000- 237	2022	- 001	- 01

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

General Electric – Boiling Water Reactor, 2957 megawatts thermal rated core power

Energy Industry Identification System (EIS) codes are identified in the text as [XX].

A. CONDITIONS PRIOR TO EVENT

Unit: 2 Event Date: July 12, 2022 Event Time: 0803 CDT

Reactor Mode: 1 Mode Name: Power Operation Power Level: 100%

B. DESCRIPTION OF EVENT

On July 12, 2022, at 0759 CDT, one of two Feedwater [JB] regulating valves (FWRV) [FCV], which are used to control feedwater flow to the reactor vessel, unexpectedly failed to the full-open position from its initial position of 42% open, resulting in a Reactor Feedwater [SK] Pump [P] trip on low suction pressure and an automatic Reactor Recirculation [AD] runback. This condition resulted in reactor water level rising due to the inability of the Feedwater system to regulate flow.

At 0803 CDT, an automatic reactor protection system [JC] actuation occurred following a main turbine [TA] trip due to high reactor water level. Plant systems operated as expected in response to the automatic scram. All control rods [AA] inserted to their full-in position and the scram was not complicated. Reactor pressure was maintained in the normal control band. Reactor water level was restored and maintained in the normal control band.

The NRC was notified via ENS Report 55988 at 1147 EDT.

This event is being reported in accordance with 10 CFR 50.73(a)(2)(iv)(A), an event that resulted in automatic actuation of a system listed in paragraph 10 CFR 50.73(a)(2)(iv)(B), specifically the reactor protection system including reactor scram.

C. CAUSE OF EVENT

The cause of the FWRV failure was due to failure of the respective air solenoid valve [SOL], which is used to control FWRV position. The solenoid valve failure prevented the FWRV from repositioning.

Failure analysis and root cause of the air solenoid valve failure was due to a manufacturing byproduct on the pilot valve plunger, which prevented the pilot valve from seating. A small piece of plastic was found on the internal seat surface. The plastic is the same material as the valve body construction. This material is not consistent with any other component in the instrument air system. Evidence supports the conclusion that the material was present when the valve was assembled at the factory.

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NARRATIVE**D. SAFETY ANALYSIS**

The safety significance of this event was determined to be minimal based upon the availability of the required systems to

- 1) Shutdown the reactor and maintain it in a safe shutdown condition,
- 2) Remove residual heat,
- 3) Control the release of radioactive material, or
- 4) Mitigate the consequences of an accident.

There were no safety consequences impacting plant or public safety as a result of this event. There was no loss of safety function for this event.

E. CORRECTIVE ACTIONS

Corrective actions included replacement of the failed air solenoid valve.

Additional corrective action(s) include inspecting similar incoming valves and certifying the valves are free of foreign material by a second or third party vendor and the solenoid valves for the feedwater regulating valves were reclassified as single point vulnerabilities to develop mitigation and elimination strategies for potential failures.

F. PREVIOUS OCCURRENCES

No previous occurrences of this event were determined through the investigation.

G. COMPONENT FAILURE DATA

Device	Manufacturer	Model	S/N	Comp Type
Solenoid Valve	ASCO	542 02 020	N/A	SOL